Algorithmic & information monocultures

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AI slop


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Made it with my own hands! 😊
Thanks to everyone who appreciates this ❤️🙏
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AI brings convergence

Algorithmic pricing under pressure: DOJ's RealPage settlement changes the rules for rental markets

Deep Learning and Financial Stability

45 Pages • Posted: 13 Nov 2020

Gary Gensler
Massachusetts Institute of Technology (MIT) - Sloan School of Management

Lily Bailey
Massachusetts Institute of Technology (MIT) - Electrical Engineering and Computer Science

How AI can undermine financial stability

Jon Danielsson, Andreas Uthemann / 22 Jan 2024

Monoculture as an analytic lens

Some critical features of algorithms & AI:

- **Scalable** (used by many people)
- **Flexible** (used in many contexts [esp. gen AI])
- **Consistent** (repeated use gives similar outputs)
- **Smooth** (small changes in use/development → small changes in outputs)

The confluence of these features is **monoculture**: previously independent entities now behave in coordinated/correlated ways.

Analyzing monoculture

Is convergence inherently bad?

What are the downstream consequences?

How will people/institutions/the world adapt?

This set of questions has been extremely generative for me.

Today: An econ/CS perspective on monoculture

- What do we know?
- How do we apply this lens?
  - Theory
  - Empirics
- Some examples and vague ideas

The value of modeling

- AI & information filtering are in flux. We can’t run experiments on what the technology will look like in 5 years.
- We want insights that will continue to hold over time
- Models abstract away from today’s capabilities & give us a view of where things are headed

Monoculture in decision-making

How do shared information structures impact decisions?

My route into this

Accordingly, you agree that you will:

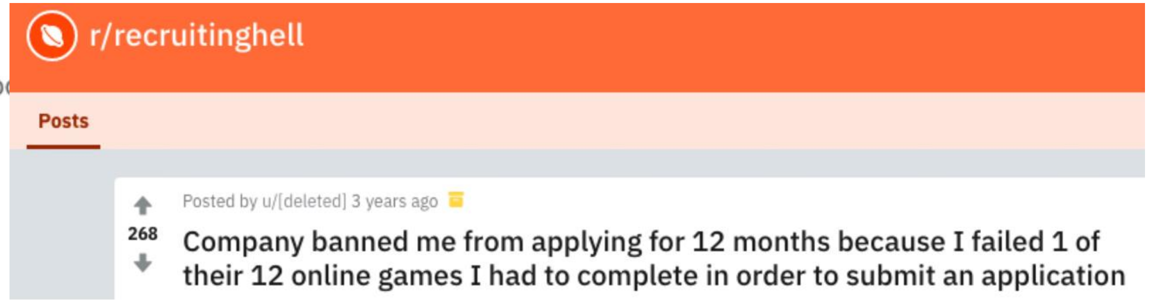
a. Take the Games seriously and play the Games attentively;

b. Carefully read the instructions before commencing each of the Games;

c. NOT stop a Game halfway through, or purposely end gameplay to repeat a particular Game;

d. NOT attempt to play any Game more than once in a 12-month period, either through your original Account or through the creation of an additional Account; and

e. NOT attempt to delete your Gameplay Data or Trait Profile for a perio



Seems frustrating. (Lots of things are frustrating.) Is there something more here?

The welfare argument

Is monoculture bad for firms (& society)?

Model: Firms sequentially choose their favorite (remaining) candidate from the same pool. Choose between common vs. independent ranking technologies.

Theorem: At equilibrium, firms may choose the common technology & get less utility [Kleinberg & Raghavan, 2021].

The welfare argument

- Let’s say there’s a common algorithm that is more accurate than any independent decision-maker
- If I choose to use it, that increases my utility
- If my competitor chooses to use it, that increases their utility **but decreases my utility**
- This is an **information externality**

An information perspective

- Each decision-maker is also contributing information to the system
- Independent decision-makers each contribute new information
- Correlated/identical decision-makers contribute less (marginal) information
- Connections to “wisdom of the crowds,” free-riding, collective intelligence.

Monoculture in matching markets [Peng & Garg 2023; 2024]

- Model: Firms have noisy preferences over applicants; applicants have uniform preferences over firms
- Stable matching solution concept
- Result: In the limit, firms with independent errors make collectively better decisions
- **Wisdom of the crowds** effect: The system works better if everyone contributes new information

Information externalities & markets

- The “externalities” framing suggests that markets should limit negative consequences
  - Intuition: If new information is valuable, then people will be incentivized to produce it
- In a model where decision-makers can choose between information sources with varying dependence, the price of anarchy is 2 [Kleinberg et al., 2026].
  - Intuition: If monoculture is bad for everyone, then some individual should benefit from switching to an independent strategy.

Status so far

- Monoculture: Shared algorithms/information lead to worse decision-making
  - Negative impacts are bounded due to market forces
- Where do we go from here?
  - Structured applications beyond decision-making
  - Unstructured settings

Monoculture in structured settings

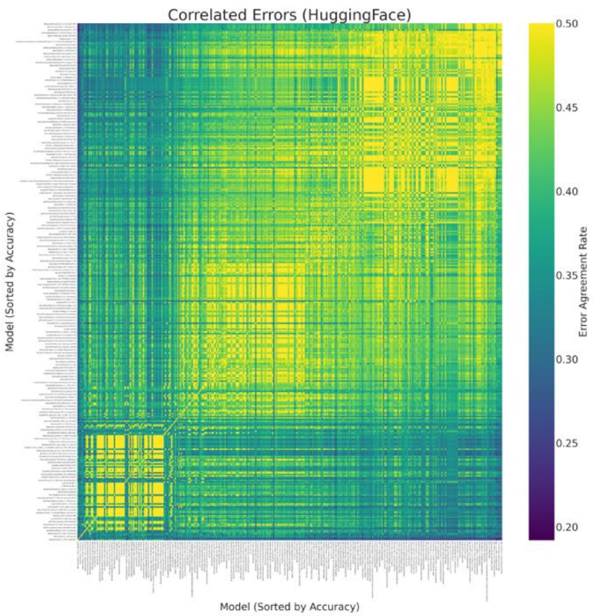
What is the dependence structure of errors?

A general lens: Correlated errors

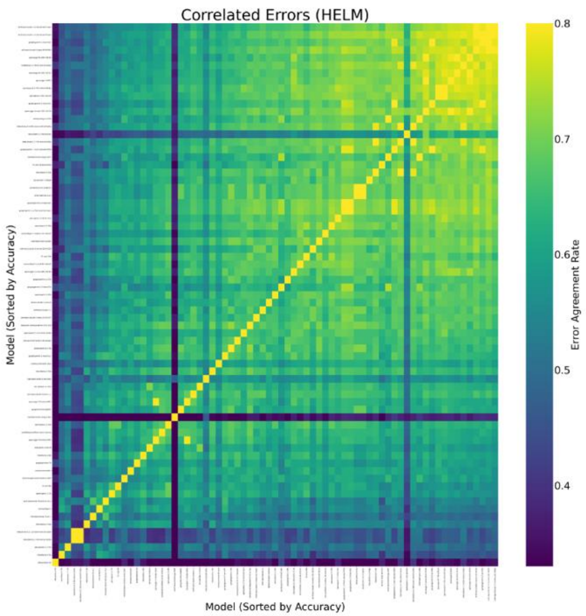
- “Outcome homogenization” [Bommasani et al., 2022]: Even if we don’t use identical models, if they have shared components/data, they will make correlated mistakes.
- What impacts might those correlated mistakes have?
- Where do you see this happening in practice?

Benchmarks

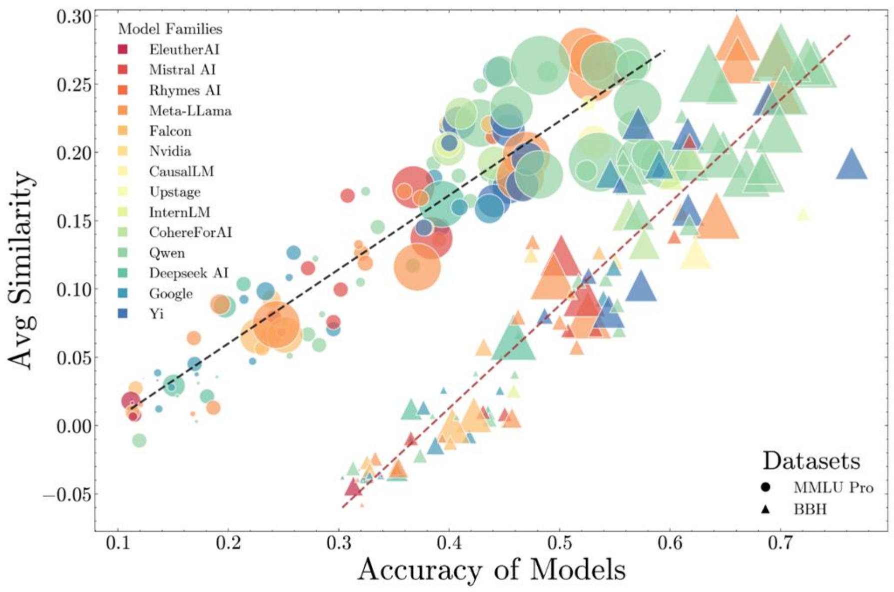
Multiple-choice benchmarks [Kim et al., 2025; Goel et al., 2025; Jo et al., 2026]: Models make correlated mistakes, and they seem to be getting more correlated



(a) HuggingFace

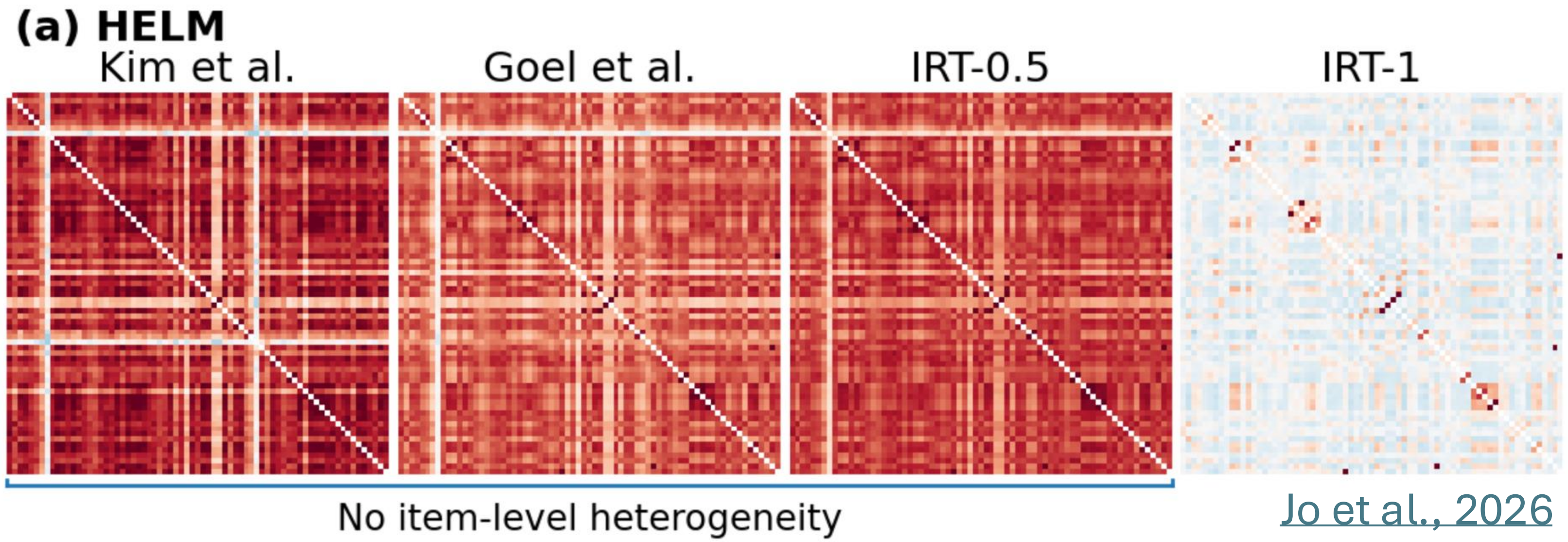


(b) Helm



Caveat: “Excess agreement” is relative

- “Models agree more than you would expect”
- What do you expect? I.e., what’s your “null model”?
    - Independent Bernoulli with fixed bias for each LLM-item pair? But some LLMs are better than others!
    - Independent Bernoulli with model-dependent bias? But some items are “harder” than others!
    - …

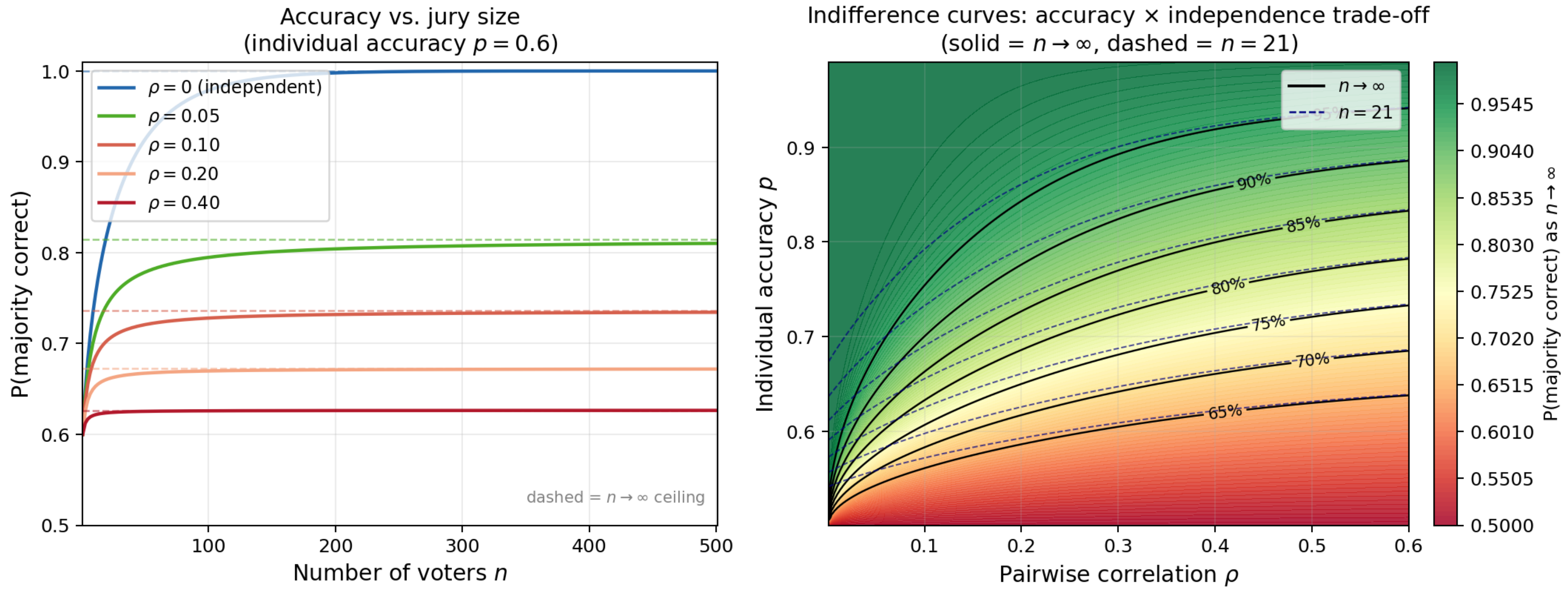


Jo et al., 2026

Voting

- Condorcet jury thm: Majority vote among independent voters is more accurate
    - “Independent” is doing a lot of work here!
- More accurate voters are better; so are “more independent” voters

Correlated Condorcet Jury Theorem (Beta-Binomial model)



Algorithmic collusion

- Oct 2022: ProPublica publishes an article making the following argument:
  - RealPage, a company that offers property management software, is used throughout the market
  - They can coordinate price increases across property managers
- Subsequent lawsuits, settlements
- Research ideas?

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Rent Barons

Rent Going Up? One Company's Algorithm Could Be Why.

Texas-based RealPage's YieldStar software helps landlords set prices for apartments across the U.S. With rents soaring, critics are concerned that the company's proprietary algorithm is hurting competition.

Algorithmic collusion

- Oligopoly model with correlated signal/error structure [Jo et al., 2025]
  - No explicit communication!
- Firms estimate WTP & price accordingly
- Correlated errors allow you to price more aggressively: Less risk of being undercut, since your competitors will price similarly
- Incentives for model production: Rational firms should favor bias over variance because low variance $\Longrightarrow$ high correlation

Market structure & adverse selection

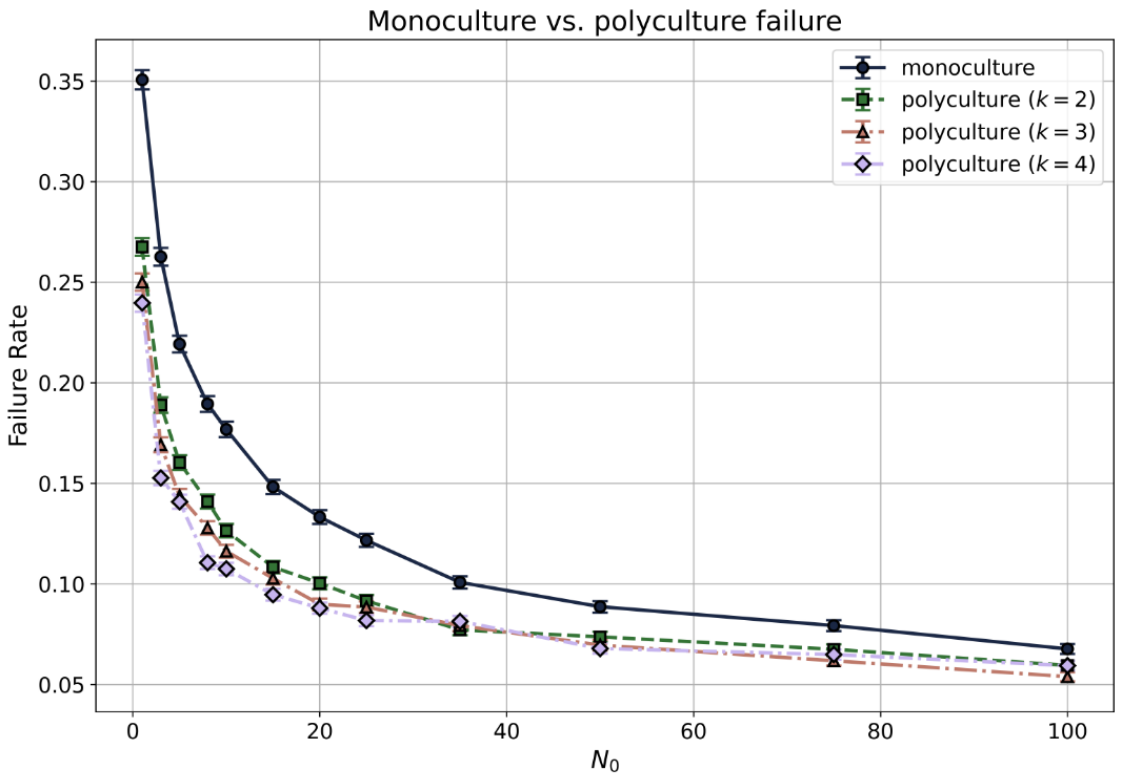
- Credit market model with correlated ("nested") screening technology [Farboodi et al., 2025]
  - If a "worse" lender screens an applicant as creditworthy, so does every "better" lender
- Questions they ask:
  - Equilibrium: Who serves whom? Rates?
  - Policy: Financial inclusion, data-sharing, etc.
- Results:
  - Higher borrower opacity $\Longrightarrow$ higher rates
  - Data sharing/transparency can help or harm

Social learning

- Models of collective information acquisition
  - Sequential social learning [Banerjee, 1992; Bikhchandani et al., 1992]: make decisions based on private information & prior observed actions
  - Bandit social learning [Banihashem et al. 2025]: make decisions based on prior actions & rewards
- A common theme: shared information inhibits social learning [Hedden & Raghavan, 2026]
  - Herding: prior observations outweigh private information
  - Exploration failures: no individual benefits from trying new, risky options

Bandit social learning

- 2 arms; sequential greedy choices; bandit feedback
- Non-zero probability of converging on "wrong" arm
- Monoculture reduces "natural" exploration, social learning
  [Hedden & Raghavan, 2026]
- Shared information is efficient in the short term but inefficient in the long term



Monoculture in structured settings

- A general playbook: Look for places in {research, your experience, the world} where independence is a load-bearing assumption
- Ask: What happens when we lose independence?
- Useful toolkit:
  - Game theory
  - Decision theory
  - Information theory
  - Probability (concentration, joint laws, ...)
  - Sequential decision-making

Monoculture in unstructured settings

What does correlation even mean?

What about Generative AI?

That Puerto Rico song going viral on TikTok is AI-generated

The catchy audio about San Juan has been shared across the platform thousands of times, despite its creator using Suno to make it.

By Matt Mitchell | May 14, 2026 | 11:39am

Andy Nairn: AI won't save creativity if we stop using our brains

Shaping the Future of Science

We believe that AI will fundamentally change how science is done across disciplines. Let us start by generating and sharing research ideas!

We are running weekly competitions where the community votes on which research ideas get implemented by AI research agents.

What about Generative AI?

- Formal correlation seems to require structured outputs. What does it mean to say that LLMs are “correlated” with each other?
- Are LLMs in “correlated” in practice? How will that change?
- What sorts of applications should we be thinking about?

What’s the mechanism here?

- For structured outputs, we seem to believe models are “smooth”
  - More shared data → more correlated errors
  - Same base foundation model → more correlated errors
- (Why) are LLMs “smooth”?
  - Two different users have slightly different prompts. Do they get similar responses?
  - Two different companies pre-train LLMs using some common data. Do those models behave similarly?

Measuring similarity

- Structured settings: exact equality suffices
- With high-dimensional data, exact equality is both unlikely & insufficient
  - $\Pr[f_1(x) = f_2(x) \approx 0]$
  - Even if $f_1(x) \neq f_2(x)$, results may be qualitatively similar
- What should we do instead?

2 approaches

Create discrete structure

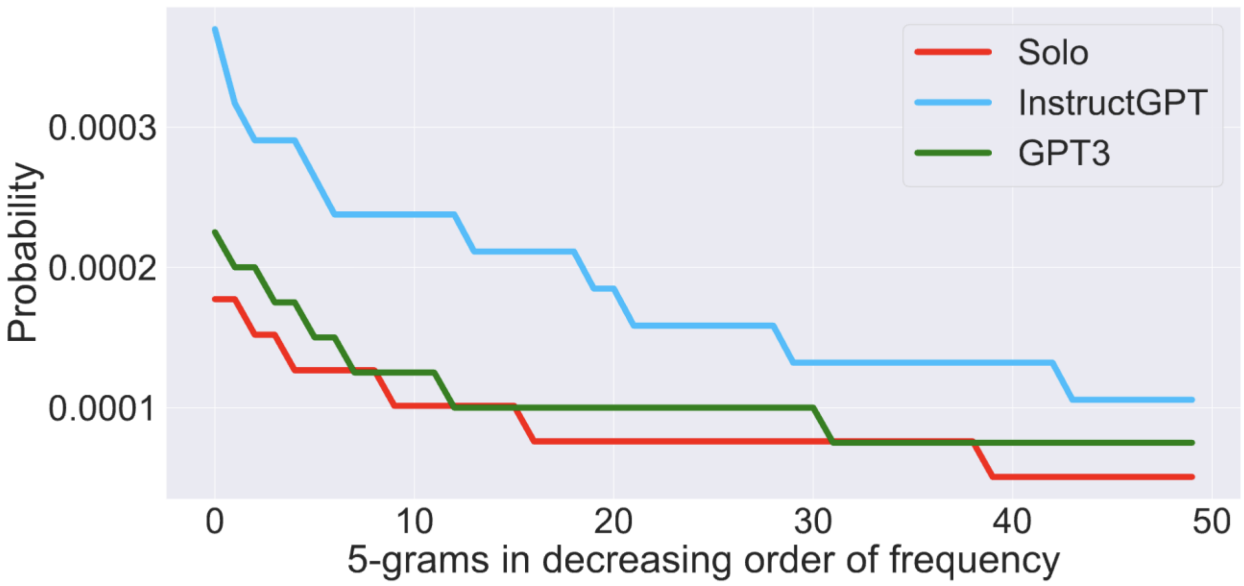
- Categorize high-dimensional data
  - Occurrence of specific words/phrases
  - Clustering
  - Extract specific info about style, substance
- Analyze collisions

Create geometric structure

- Put data into a space where geometry is meaningful
  - I.e., embeddings
- Measure similarity in that space

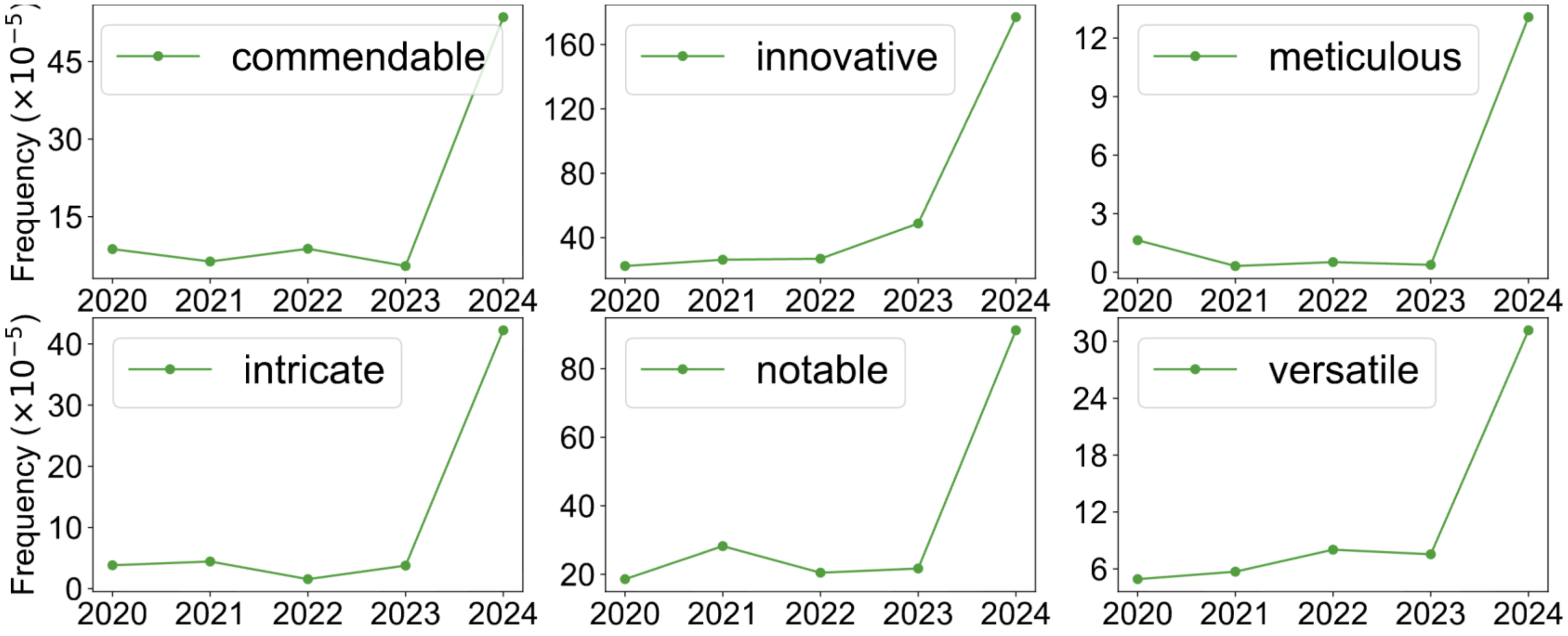
Discrete structure: AI & writing

- RCT where some participants get GPT, others don't [Padmakumar & He, 2024]
- How similar is their writing? Look at distribution of 5–word strings



Discrete structure: AI & writing

- Indicators of AI use in peer review: LLMs use certain words much more often than people [Liang et al., 2026]



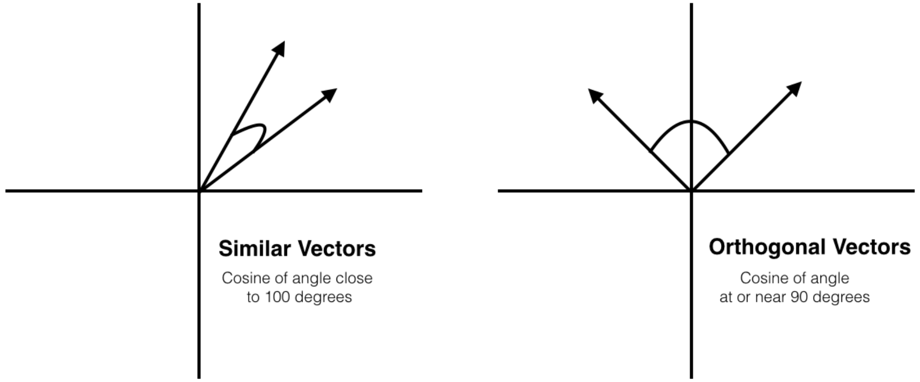
Geometric structure: Embeddings

This gives us a way to say "Condition A is more correlated than condition B"

- $E_{x,x' \sim A}[x^T x] \ll E_{x,x' \sim B}[x^T x]$

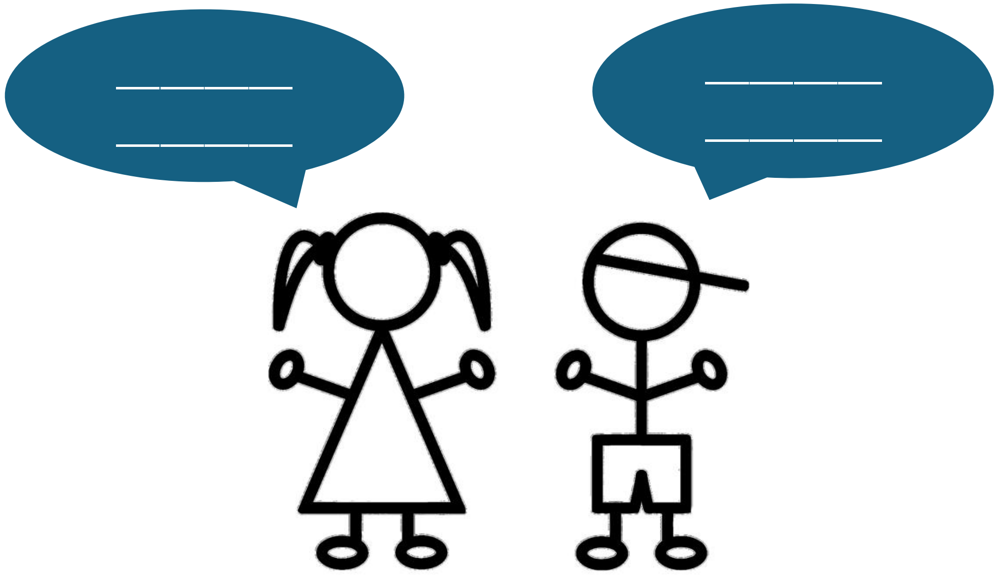
Limitations:

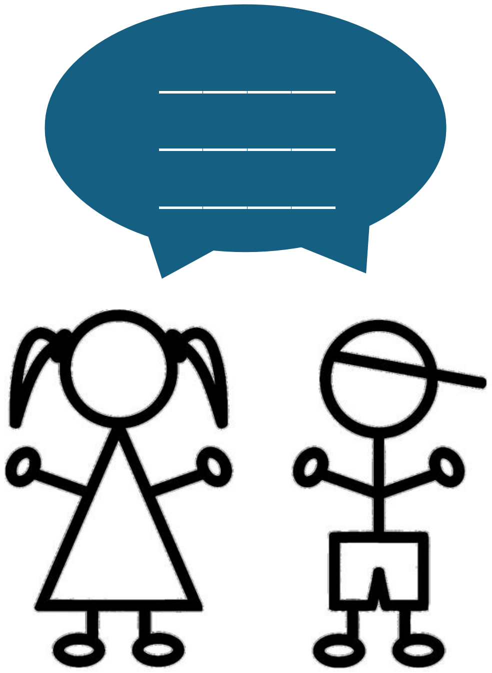
- Embeddings conflate lots of things: style, substance, …
- Harder to do theory with (more on this later)
- Empirics reliant on some black box



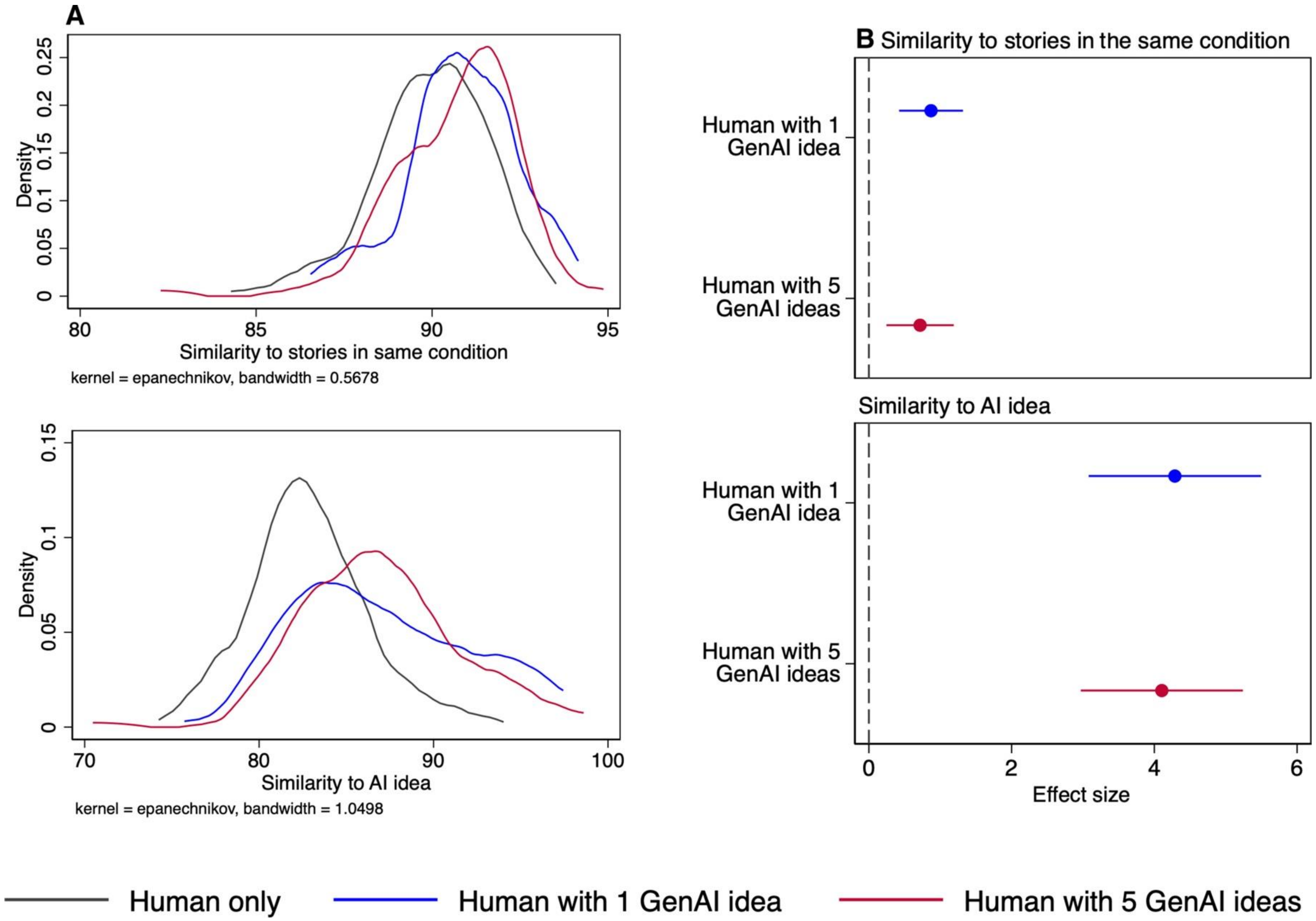
Some empirical evidence: AI & creativity

RCTs on human-AI creativity [e.g., Anderson et al. 2024; Doshi & Hauser, 2024]: Individual gains, collective losses



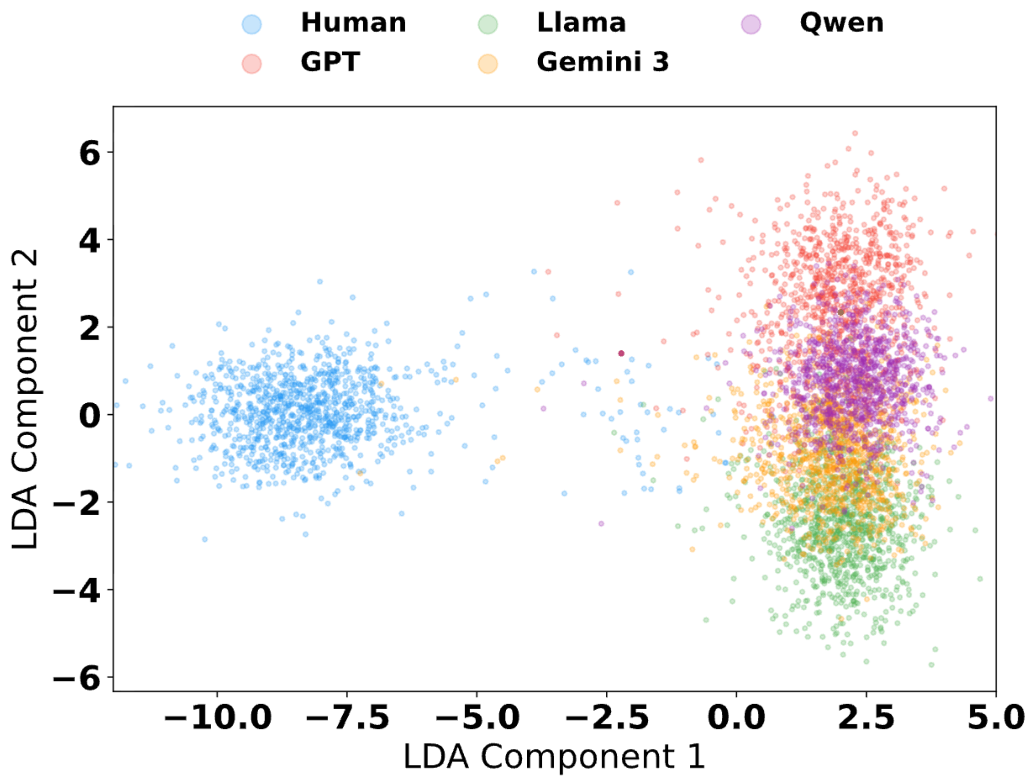


Embeddings converge



AI writing detection

- Current approaches to AI detection: Supervised ML
  - Bad for several reasons: Adversarial behavior ("humanizers"), distribution shift, different LLMs, ...
- Potential alternative: Semi-supervised methods to leverage similarity of AI-written text [Ren et al. 2026]

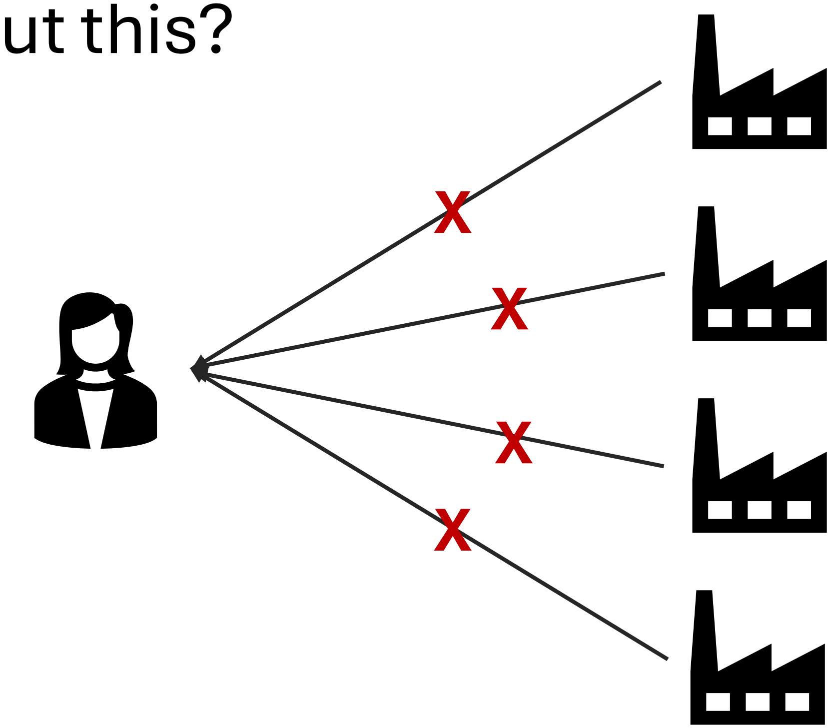


Normative aspects of monoculture

What makes monoculture good or bad?

Systemic exclusion

- Claim: Monoculture in decision-making is bad because it systematically excludes some individuals [Ajunwa, 2021; Creel, 2026]
- What do you think about this?



Exclusion & externalities

- Decision contexts matter [Hedden & Raghavan, 2026]
  - Employment: Monoculture probably won't change unemployment
  - Awards: Monoculture (one person gets all the awards) is likely bad. What's the point of having multiple awards if they all go to the same person?
- Is randomization desirable? [Creel, 2026]
  - May lead to less inequality
  - Who wants to randomize?

Agency

- Will monoculture change the distribution of effort?
  - Specialization
  - Give up? Try harder?
- Interesting modeling opportunity

Looking forward

Is this a persistent technical phenomenon?

Practical considerations

- In many applications, consistency is desirable
  - E.g., in many business applications, you don't want unpredictability
- If these are the highest-value applications, we should expect general-purpose tools to be consistent by design
  - There's a growing body of literature on making AI **more** consistent [e.g., Elazar et al. 2021]
- Generating high-quality random variation is hard (more on this tomorrow)

Personalization

- Communication bottleneck: Even if you have rich preferences, it's not worth communicating them [Castro et al. 2023; Park et al. 2026]
- Better personalization may mitigate monoculture
- Related to the literature on "steering" [e.g., Vafa et al. 2025]

Technical obstacles

- Existing training methods lead to “mode collapse” [c.f. Goodfellow et al. 2014]
- RLHF is miscalibrated [Slocum et al. 2025]
  - Maximize {likelihood of answer} – {regularization term}
  - Regularization is not a proper scoring rule
  - $\pi(y) \propto \pi_{\mathrm{ref}} \cdot p^{1/\beta}$
- Distributional alignment is hard!

Closing thoughts

- Monoculture is a broad & useful analytic lens
  - Mix of CS, econ, law, psychology, …
  - Basic structure: Shared information → shared behavior
- Some generative questions
  - Where do we (implicitly) rely on independence?
  - Do market forces encourage divergence?
  - What is the interaction between quality/efficiency & correlation?

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